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Wavelet Entropy and Support Vector Machine Multi-Class Fall Detection Based on IMU Wearable Sensors Data

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***Abstract*—Fall detection is especially important for seniors because falls are one of the main health dangers for the elderly in public health care. Elderly people are particularly vulnerable to falls, which can result in serious injuries, especially if the person who falls remains on the ground for an extended period without help. In this paper, we proposed a fall detection system-based Inertial Measurement Unit (IMU) sensor to improve the detection method. Wavelet entropy for feature extraction and a directed acyclic graph support vector machine (DAG-SVM) classifier for multi-class classification are then applied. The proposed system is evaluated on a ready-made dataset based on multi-wearable sensors; data come from multi-sensors chest, thigh, and waist IMU datasets. The multi-class classification approach splits the elderly activities into three phases: pre-fall, impact, and post-fall. Experimental results, relying on using a single accuracy in fall detection were compared with multiple sensors and achieved 71.66%, 71.64%, and 97.79%, with different sensor locations. The results showed that the proposed model-based multi-wearable sensors demonstrated high accuracy compared to the model-based single sensor.**

Keywords—fall detection, IMU wearable sensors, entropy wavelet, Directed Acyclic Graph SVM, single sensor, multiple sensors

I. INTRODUCTION

The health and quality of life of the elderly have become increasingly important as the global population grows. The World Health Organization (WHO) projects that between 28% and 35% of those 65 years of age and older may experience falls each year. The risk rises for persons 70 years of age or older from 32% to 42% [1]. Furthermore, more than 50% of individuals who suffer injuries show a notable death rate within six months after the accident [2]. Using wearable sensors to distinguish falls from traditional Activities of Daily Living (ADLs) presents several obstacles, mostly related to the complexity and variety of human movements, the requirement for efficient detection algorithms, and the scarcity of realistic fall data for training and evaluation purposes.

Fall detection in the elderly is a critical aspect of healthcare and assisted living, aiming to provide timely assistance and mitigate the potential consequences of falls. Wearable sensors, such as Inertial Measurement Units (IMUs) and accelerometers, offer a valuable means of capturing continuous and real-time data on the movement patterns of individuals. Numerous studies on the fall detection system have been conducted. There are no set standards based on sensors, computed characteristics, or classification methods. To detect falls, images from a camera are captured and then processed using image processing techniques. Fall and ADL are the main focus of the majority of fall detection tests. When it comes to fall detection, falls can be categorized into four phases: pre-impact, impact, unstable, and stable. Numerous studies have described falls using the properties of the impact and stable phases [3]. Post-impact fall detection is a form of fall detection technique that may identify whether elderly individuals have fallen but are unable to stop falls from injuring them. Alternatively, pre-impact fall detection attempts to find the moment before fall. An airbag system or inflatable hip protection can be deployed once a pre-impact fall is detected, greatly reducing damage for the elderly [4-5]. However, because the free-fall phase is unclear, pre-impact fall detection is still a difficult problem. Few research has used wearable IMU sensors as experimental equipment for pre-impact fall detection. [4],[6]. Wearable sensors like an accelerometer, and gyroscope at the knee, wrist, neck, and waist [7],[8][9][10], [11], [12],[13], [14] are used to get the data input. Wearable sensors, integrated into devices like smartwatches [8] or body-worn accelerometers, provide a wealth of information about an individual's movements, postures, and activities. These sensors offer the advantage of continuous monitoring, enabling the detection of subtle changes in movement patterns that may precede a fall. The rich time series data generated by wearable sensors serves as a valuable resource for training machine learning models capable of discerning complex patterns associated with falls.

Sensor readings are used to construct features, which extract useful information from the raw data. Moreover, ambient sensors like infrared or movement-based sensors can be used to identify falls.

The patient-specific fall prediction and detection system described in [15] utilizes a single tri-axial accelerometer sensor attached to the patient's thigh. The performance of the system has been validated using the MobiFall Dataset and experiments. The fast mode achieves a sensitivity of 97.8% and specificity of 99.1%, while the slow mode achieves a

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sensitivity of 98.6% and specificity of 99.3% for a total of 600 measured ADL and falls from 77 subjects [15].

Fall detection is a critical component of elderly care to ensure prompt response and minimize the consequences of falls. Traditional fall detection systems often rely on simple threshold-based approaches, which may lack the ability to capture the complex temporal patterns leading to a fall. Therefore. this work proposes a fall detection model that analyzes two public datasets obtained from chest, waist, and ankle, sensors using Structured Support Vector Machines (SSVMs) to distinguish between falls and activities of daily living. We use a readymade dataset called inertial measurement units (IMU) [14]. The SSVMs offer a powerful solution by considering the temporal dependencies and sequential nature of time series sensor data. Studies that have already been done have also contrasted the performance within each dataset. The approach and support vector machine-directed acyclic graph support vector machine methods are presented in Section 2 and will be used throughout the study. The dataset utilized for verification is described in Section 3, and the model evaluation's findings are shown in Section 4. Ultimately, the study's conclusions are presented in Section 5.

II. Methodology

In this work, we propose a fall detection model and hybrid method for multi-class classification to improve fall detection accuracy. Fig.1 shows the proposed fall detection model. A multi-class approach in fall detection refers to the use of machine learning algorithms to classify falls into multiple categories or phases, rather than simply detecting whether a fall has occurred or not. This approach involves identifying different phases of a fall, such as pre-fall, daily activity living (no-fall), and fall phases, and classifying them into distinct classes. By differentiating between these phases, the multi-class approach aims to provide more detailed and nuanced information about the nature of a fall event, which can be valuable for understanding the circumstances surrounding the fall and providing appropriate assistance or intervention. The multi-class approach in fall detection represents a more advanced and detailed method for analyzing and categorizing fall events, with the potential to improve the safety of those who are in danger of falling and increase the efficacy of fall detection systems. In our method, we used the entropy wavelet technique for feature selection and then applied a Directed Acyclic Graph SVM (DAG-SVM), and a publicity dataset namely the IMU dataset [14] was used to evaluate the method.

Wavelet Entropy (WE) and Directed Acyclic Graph Support Vector Machine (DAG-SVM) work together in fall detection for the elderly as follows: feature extraction using wavelet entropy: wavelet Entropy is utilized to extract features from sensors data obtained through IMU sensors sat on the waist, chest, and ankle. The extracted WE features capture important patterns and characteristics related to fall and non-fall activities; then the extracted WE features are input into the Directed Acyclic Graph Support Vector Machine (DAG-SVM) for classification. DAG-SVM is a variant of the traditional Support Vector Machine (SVM) that extends its capability to handle multi-class classification problems by using a directed acyclic graph structure. This allows for more complex decision boundaries and improved classification performance in multi-class scenarios.

A. Wavelet Entropy

Wavelet Entropy (WE) is a method used to extract global features from signals [16], such as sensor data obtained from IMU sensors. The method involves two main steps: discrete wavelet decomposition and entropy computation. Fig. 3 shows level 2 of the IMU dataset using bior5.5 and Daubechies functions.

. In the first step, the discrete wavelet decomposition was carried out in the first stage on the IMU dataset for three sensors (thigh, waist, and ankle), three sub-bands (details coefficient level 1, details coefficient level 2), and approximation coefficient), were yielded. Here approximation and details represent the low- and high-frequency coefficients. n+1 sub-band are obtained for an n-level decomposition. The second step involved computing entropy for every sub-band. 256 x 256 sensor data can be reduced to a (n+1)-element vector by the WE in total. The WE pseudocode is provided in Table 1 below. Table 1 indicates that to perform a WE, two parameters are required: the wavelet family and the decomposition level.

A discrete wavelet transform is used to break the signal down into several frequency sub-bands. The wavelet transform is a mathematical technique that decomposes a signal into different frequency components, allowing for the analysis of the signal at different scales [16]. The decomposition results in a set of sub-bands, each representing a different frequency range of the signal. In the second step, entropy is calculated over each sub-band. Entropy is a measure of the amount of disorder or randomness in a signal. Quantifying the distribution of the wavelet coefficients in each sub-band is necessary for the entropy computation. The entropy value provides a measure of the complexity or irregularity of the signal in each sub-band. The WE method combines the entropy values from all sub-bands to create a feature vector that captures the global characteristics of the signal. The feature vector can then be used for classification. The method's ability to capture complex patterns and characteristics in signals makes it a valuable tool for feature extraction in various domains, including medical imaging, speech processing, and financial analysis.

B. Wavelet Family

Wavelet families refer to different sets of wavelet functions that are used in wavelet analysis for signal processing and data compression. Each wavelet family has specific properties that make it suitable for different types of signals and applications. Different wavelet families may be more suitable for capturing certain features or patterns in the data. We used the bior5.5 wavelet family [17] for decomposition, which is a specific case of biorthogonal wavelet pairs with desirable properties for analyzing fall and daily activity data.

TABLE I. Wavelet entropy pseudocode. Wavelet entropy (WE) as an algorithm

Step 1 Bring the sensor's data in.
Step 2 Select decomposition level n and wavelet family
Step 3 Decomposition and generate (n+1) sub-bands.
Step 4 Determine the entropy for every sub-band.
Step 5 As a column vector, combine all of the entropy results and output it as the feature.

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C. *Support Vector Machine (SVM) and Directed Acyclic Graph-SVM (DAG-SVM)*

A one-on-one strategy can be used to expand the binary classifier known as the support vector machine classifier to handle multi-class classification [18]. SVM has been applied in the context of multi-class fall detection to differentiate between different phases of a fall event. SVM is a machine learning approach based on statistical learning theory. It was first presented for binary classification [18]. Since real data sets of multi-class are widespread, much work has been done on how to extend binary SVMs to multi-class circumstances, such as the one-against-one and directed acyclic graph (DAG) techniques [19].

As shown in Fig. 2, the classifier is broken down into three binary classification problems with the use of DAG. Three classifiers are established here are: ADL-vs-Fall, ADL-vs-PF, and Fall-vs-PF. Next, the three classifiers are connected using the DAG-SVM method. The DAG-SVM is a variation of SVM that is designed to handle multi-class classification problems [20]. DAG-SVM constructs N(N-1)/2 classifiers [21], one for each pair of classes, where N is the number of classes. The approach uses two-class maximal margin hyper-planes at each decision node of the Decision Directed Acyclic Graph (DDAG), operating in a kernel-induced feature space. Training each ij node solely on the subset of training points designated by i or j yields the DDAG, $(i=1,2,3.,...N-1, j=1,2,3,..N)$. The DDAG tests the first and last elements of the new list (x) until only one class is left in the list. It then obtains the score of L_{ij} of ij th individual classifier, whose output is the sign function of the score value L_{ij}. A test point is evaluated against the decision node that corresponds to the first and last elements of the list [22].

$$o_{ij(x)} = \mathrm{sgn}(L_{ij(x)}) _____ (1)$$

If the score $L_{ij}(x)$ is larger than zero, then the output O_{ij} (x) is +1, denoting that x does not belong to jth class; otherwise, the output is −1, denoting x does not belong to ith class [23]. The directed Acyclic Graph Support Vector Machine (DAG-SVM) is employed to handle the multi-class problem of differentiation.

The significance of using a directed acyclic graph support vector machine (DAG-SVM) in distinguishing human postures and detecting fall-like accidents lies in its ability to effectively handle multi-class classification problems and provide an excellent balance between generalization error and time efficiency. Fall, pre-fall, and ADL postures are the three categories into which the DAG-SVM is used to classify postures that are closely associated with fall event detection, particularly the laying postures. As shown in Fig. 2, it makes use of a tree-like structure in which each node is equivalent to a simple binary SVM. When compared to existing multi-class SVM methods, this method offers an ideal compromise between generalization error and time efficiency at both training and compute times by combining multiple binary classifiers to address multi-class classification problems. the DAG-SVM plays a crucial role in effectively classifying human postures and detecting fall-like accidents, making it a significant component of the proposed fall detection approach.

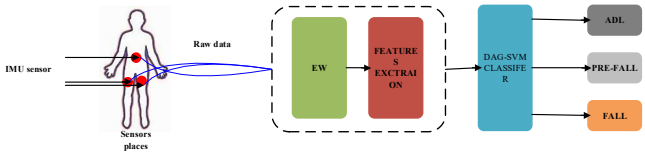


Fig. 1. Diagram of fall detection model

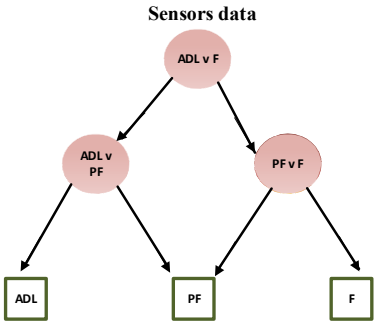


Fig. 2. Diagram of the DAG-SVM for the human activity classification (fall (F), non-fall (DAL), pre-fall (PF))

III. DATASET

In this section, we discuss waist-based, thigh-based, and ankle-based fall-based datasets for validating the fall detection algorithm. the waist-based open dataset includes the SisFall dataset [10], FallAllD [12], and others, and Thigh-based open dataset includes the MobiFall dataset [11] and. and waits, thigh, ankle-based dataset IMU dataset [14], UP-Fall [8], and UMA-Fall [24] and others. In this study, IMU was selected as a dataset for verification because IMU possessed the largest amount of data, and contained fall, pre-fall, and ADL activities.

A. *IMU dataset*

This dataset consisted of body kinematics data recorded using an array of seven tri-axial accelerometers (range ±6 g, Opal model, APDM Inc., Portland, OR). The accelerometers were mounted bilaterally on the ankles and thighs, as well as at the anterior aspect of the waist, sternum, and head. Data were recorded at a sampling rate of 128 Hz for 15 seconds per trial [14]. The study comprised ten young adults in good health, ages ranging from 22 to 32. The mean age of the participants was 26.6 years, with a standard deviation of 2.8 years. All participants were Simon Fraser University students, who were found through fliers and ads posted on campus bulletin boards.

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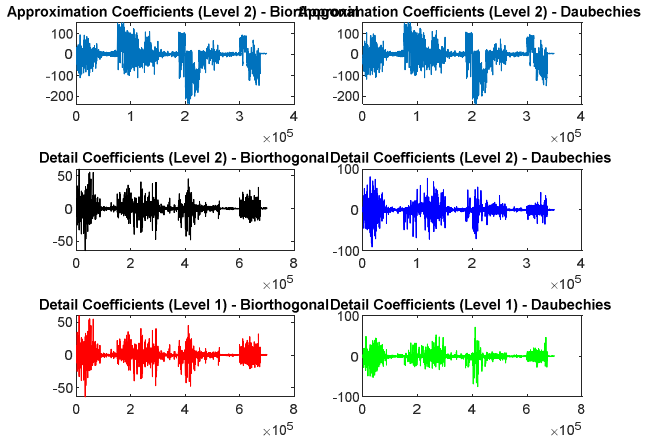


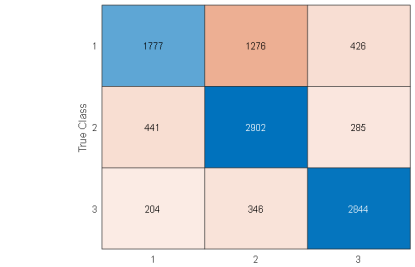
Fig. 3. Decomposition for IMU Dataset using bior5.5 and Daubechies functions.

B. *Optimal Decomposition Level*

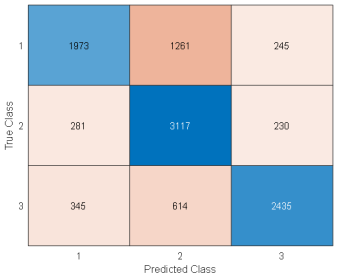
Determining the optimal decomposition level in Wavelet Entropy (WE) involves finding the level of wavelet decomposition that provides the best performance for the specific task or dataset at hand. The optimal decomposition level is typically selected based on criteria such as classification accuracy, information content, or other performance metrics.

IV. Results and Discussions

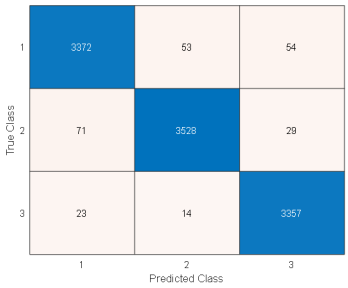
The experiments were run using MATLAB 2022a software and the system requirements for running different iterations are an Intel Core i5 processor, a 64-bit processor, a single CPU, and 32 GB of RAM. The performance of prediction and detection fall for elderly people is investigated using evaluation measures namely accuracy, precision, recall (sensitivity), specificity, and F-score on all datasets. We divided the dataset into two: 80% for training and the remaining 20% for testing for all datasets. DAG-SVM was trained each time and the confusion matrices over the IMU dataset are shown in Fig. 4.



a.



b.



c.

Fig. 4. Confusion matrix for IMU dataset USING ADG-SVM. a. Thigh b. Waist c. Waist, Thigh, and ankle

Lastly, a comparison was carried out with alternative method aids for evaluating the efficacy of the DAGSVM methodology. Table II and III show the comparison of the DAG-SVM compared to space of art and the accuracy of the DAG-SVM is 97.79%, higher compared to SVM and KNN. When using three sensors, the results showed that the accuracy was higher in all algorithms proposed, i.e., DAG-SVM, SVM, and KNN, highlighting the advantage of DAG-SVM over single-sensor approaches in evaluating fall risk due to its capacity to leverage richer and more diverse information.

TABLE II. Fall detection results for the IMU dataset Multiple Sensors

Sensor location	ML	Accuracy	specificity	Precision	Sensitivity	f-score
Waist, thigh, and ankle	SVM	**95.9%**	95%	94.8%	94%	95.83%
	KNN	**96.3%**	95.6%	94.8%	96.4%	96%
	DAG-SVM	**97.79%**	97.45%	97.3%	97.38%	97.34%

TABLE III. Fall detection results for the IMU dataset Single Sensor

Sensor location	ML	Accuracy	specificity	Precision	Sensitivity	f-score
Thigh	SVM	72%	86%	76%	57%	65%
	KNN	95.89%	95%	96.5%	96%	96%
	DAG-SVM	71.66%	85.91%	75.91%	56.71%	64.92%
Waist	SVM	72%	80%	73%	51%	60%
	KNN	95.7%	95%	96%	95%	96%
	DAG-SVM	71.64%	80%	73.37%	51.07%	60.23%

V. Conclusion

In this study, we proposed a fall detection system based on an IMU wearable sensor using wavelet entropy for feature extraction and a Directed Acyclic Graph Support Vector Machine (DAG-SVM) to handle the multi-class problem to differentiate between fall, pre-fall, and ADL. The ready-made dataset was used namely the IMU dataset. The utilization of DAG-SVM demonstrates its effectiveness in handling complex classification tasks in the context and achieves an overall accuracy of 97.79% when using three sensors thigh,

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waist, and ankle; achieves an overall accuracy of 71.66% when using a single sensor sat on the thigh, and achieves overall accuracy 71.64% when used single sensor sat on waist. This high accuracy highlights the potential of the proposed approach in accurately identifying falls in the elderly using wearable sensors.

These findings contribute to the field of fall detection in the elderly using multiple wearable sensors by showcasing the potential of advanced computational techniques, such as Wavelet Entropy and Directed Acyclic Graph Support Vector Machine, in improving the accuracy and efficiency of detecting falls in the elderly using multiple wearable sensors data. While traditional fall detection methods often rely on simplistic threshold-based approaches, Structured Support Vector Machines (SSVMs) present a sophisticated and promising solution for enhancing the accuracy and reliability of fall detection systems. The combination of WE as the extracted features and DAG-SVM as the classification algorithm enables the system to effectively differentiate between different classes of fall and non-fall based on the features extracted from sensor data. In summary, Wavelet Entropy is employed to extract global features capturing important patterns from sensor data, and these features are then utilized by the Directed Acyclic Graph Support Vector Machine for accurate classification of the elderly.

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